

Wasi M.F. Naqvi

 [Personal Website](#) |  [Github](#) |  [Linkedin](#) |  wasi.naqvi@mail.mcgill.ca

Education

2026–	PhD Physics, McGill University Supervisor: Nicolas B. Cowan Thesis: A Patchwork of Worlds: Artificial Intelligence for Population-Level Studies of Exoplanet Atmospheres	GPA: 4.00/4.00
2025–26	M.Sc. Physics, McGill University Supervisor: Nicolas B. Cowan Thesis: HERMES: HiERarchical Modelling for Exoplanet Science	GPA: 4.00/4.00
2021–25	B.Sc. Physics (Honours), The University of British Columbia Minor in Data Science and Statistics Thesis: Simulating In-Orbit Performance for the CASTOR Mission Supervisor: Tyrone Woods Distinctions: Dean's List, Best Thesis Award	GPA: 4.00/4.33

Publications

Naqvi, W. M. F. and Cowan, N. B. *HERMES: HiERarchical Modelling for Exoplanet Science*. Royal Astronomical Society, Techniques and Instruments (2026, Submitted, under review.). [Available on arxiv](#)

Côté et al. (including **Naqvi, W.**). *The CASTOR Mission*. Special Issue, Journal of Astronomical Telescopes, Instruments, and Systems, SPIE. (2025)

Awards & Honours

2025–	Kharusi Family International Science Fellowship, McGill University	\$15,000 CAD
2021–25	Karen McKellin International Leader of Tomorrow Award, UBC	\$250,000 CAD
2025	McGill Graduate Excellence Award	\$5000 CAD

Contributed Talks & Posters

 Contributed Talk  Poster

 HERMES: HiERarchical Modelling for Exoplanet Science CASCA Annual General Meeting.	Jun 2026 Montréal, Canada
 Comparative climatology of exoplanets using Hierarchical Modelling, Trottier Institute for Research on Exoplanets	May 2026 Montréal, Canada
 HERMES: HiERarchical Modelling for Exoplanet Science Ariel Open Conference '26	Mar 2026 ESA Harwell, UK
 BayesBall: Bayesian Modelling for sport analytics, Trottier Space Institute Lunch Talks	Nov 2025 Montréal, Canada
 A detector simulation pipeline for the CASTOR mission (Invited) The CASTOR Mission Consortium, NRC Herzberg Astronomy & Astrophysics	Jun 2026 Victoria, Canada

Research & Work Experience

Trottier Institute for Research on Exoplanets, McGill University <i>Summer Researcher</i>	Montréal, QC May 2025–Present Hierarchical Bayesian Modelling of planetary targets to probe population-level trends for the Ariel Space Mission.
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NRC Herzberg Astronomy & Astrophysics

Junior Adaptive Optics Research Scientist

Victoria, BC
Jan 2025–Present

Implementing a convolutional neural network on an Adaptive Optics test bench for the 1.2 m REVOLT Telescope at the Dominion Astrophysical Observatory. Running machine-learning simulations alongside lab experiments under Dr. Maaïke van Kooten.

TRIUMF

Universal Polarization Research Assistant

Vancouver, BC
Sep 2024–Present

Devising a Faraday Rotation method to measure polarization of optically-pumped Rb vapour at ISAC-II. Writing numerical simulations for polarizer experiments and developing a universal laser-polarization method for isotopes with unknown atomic transitions.

NRC Herzberg Astronomy & Astrophysics

Junior Research Scientist

Victoria, BC
May–Aug 2024

Designed Python software pipelines for imaging-sensor simulations and detector modelling for the CASTOR mission (Canadian Space Agency / NRC). Built with Dask (parallel computing) and ESA Pyxel (detector physics).

COCANA Lab, UBC Okanagan

Undergraduate Research Assistant

Kelowna, BC
May–Aug 2023

Optimization research for a commercial road-building project (SoftTree Inc.) under Dr. Yves Lucet. Used YALMIP solver; translated codebases between MATLAB and Python with unit-test verification.

CAN-SBX Stratospheric Balloon Design Team

Physics & Data Science Lead

Kelowna, BC
Sep 2023–May 2024

Led physics instrumentation and data analysis. Developed efficient MicroPython / C++ firmware for flight micro-controllers, designed ML models for atmospheric prediction, and assisted with dual-polarising antenna calibration and circuit analysis.

Teaching Assistantships

PHYS 180 Space, Time and Matter

Sep-Dec 2025

PHYS 241 Signal Processing

Jan-Apr 2025

Technical Competitions

Primary role across competitions: building, packaging, and benchmarking Bayesian models, U-Nets, Transformers, and CNNs for scientific data challenges.

ExoHack: Ariel Machine Learning Hackathon (**Winner**)

Mar 2026

NeurIPS Ariel Data Challenge

Jun 2026

Rocky Worlds DDT Challenge

May 2026

Community Engagement & Outreach

Founding Member & Organizing Committee, Montréal AstroStats Day

2026–

[Author, Astrobites](#)

2025–

iREx EDI Committee Member

2025–

McGill Hackathon Mentor

2026–

Astro on Tap Volunteer

2026–

Volunteer, McGill Physics “World of STEM” high-school astronomy sessions

2025–

Coordinator, graduate student lunches with TSI seminar speaker

2025–

Facilitator, McGill “From Planets to Particles” family science outreach fair

2025–

Organizing & Administration Volunteer, Exoclines 2025 Conference

2025–

Science Outreach, Trottier Institute for Research on Exoplanets

2025–

UBCO Physics Student Club Treasurer

2021–2025

Exercise is Medicine Coordinator

2023–2025

Volunteer Soccer Coach, Royal Oak, Victoria

2024–2025

Mentor, CERN BeamLine For Schools (BL4S), Cedar College, Karachi

2021–

Science Outreach, Let's Talk Science Okanagan
Volunteer, Okanagan Food Bank
Volunteer, Student Union Okanagan, UBC

2023–2024
2022–2024
2022–2023

Skills

Programming: Python, C, C++, Julia, Fortran, JavaScript, R, SQL, Shell/Bash

Markup & Typesetting: LaTeX, HTML/CSS, Markdown

Python Ecosystem: PyTorch, JAX, NumPyro, PyMC, Emcee, NumPy, SciPy, Astropy, Scikit-learn, Dask, Matplotlib, Pandas, ArviZ, Corner, Dynesty

Statistical & ML Methods: Hierarchical Bayesian Modelling, Gaussian Processes, MCMC Sampling, Nested Sampling, Simulation-Based Inference, PCA, CNNs, U-Nets, Transformers, Variational Inference, Model Comparison and Benchmarking

N-Body Codes: REBOUND

Computing Infrastructure: CANFAR, DRAC Supercomputing Clusters, Slurm, MPI, Git/GitHub, Docker, Linux

Other Software: Microsoft Excel, Google Sheets

Relevant Coursework

Radiative Processes in Astrophysics, Statistical Learning, Computational Astrophysics, Machine Learning Theory, Probability Theory, Stochastic Processes, Exoplanets and Brown Dwarfs.